How do Organisms Reproduce?

Science • Chapter 7 • *Chapter Summary & Study Guide for Daksh*

Reproduction is the only life process not necessary for the survival of an individual — but absolutely essential for the survival of the species. It also creates the **variation** on which evolution acts. This chapter looks at the strategies different organisms use, with a special focus on flowering plants and humans.

Why reproduction creates variation

DNA copying is the basis of reproduction — but no copying is perfect, and these small differences are **variations**. In sexually reproducing organisms, two cells from two parents combine, mixing the DNA further. Variation is crucial: in a changing environment (heat, disease), some variants survive while others do not — this drives evolution.

Asexual reproduction

Only one parent; offspring are nearly identical clones. Common modes:

- **Fission** — one cell splits into two. Binary in *Amoeba*; multiple in *Plasmodium* (malaria parasite).
- **Budding** — a new individual grows out as a bud from the parent. Seen in *Hydra* and *Yeast*.
- **Fragmentation** — body breaks into fragments, each grows into a new organism (e.g. *Spirogyra*).
- **Regeneration** — a body part can grow into a whole organism (e.g. *Planaria*). Different from fragmentation: regeneration relies on specialised cells that re-organise.
- **Spore formation** — non-motile spores released into air; each germinates on a moist substrate (e.g. *Rhizopus* bread mould).
- **Vegetative propagation** in plants — new plants from roots, stems or leaves. *Examples:* potato (stem tuber 'eyes'), bryophyllum (leaf margins), rose (stem cuttings), banana (rhizome). Used to propagate seedless varieties (banana, orange).
- **Tissue culture** — many tiny new plants grown from a few cells in a lab; used commercially.

Sexual reproduction in flowering plants

The flower is the reproductive organ. Important parts:

- **Stamen** (male) — anther produces **pollen grains**; filament holds it up.
- **Pistil/Carpel** (female) — stigma (sticky), style, ovary containing **ovules** (egg cells).

Pollination = transfer of pollen from anther to stigma — by wind, water, insects, birds. Self-pollination (same plant) or cross-pollination (different plant). After landing, pollen germinates and grows a **pollen tube** down the style to the ovule, where the male gamete fuses with the egg cell — **fertilisation**. The zygote becomes the embryo, the ovule becomes the seed, and the ovary develops into the fruit.

Sexual reproduction in human beings

Puberty brings the body to reproductive maturity (~10–14 yrs in girls, ~12–16 in boys). Visible changes (secondary sexual characteristics) include hair growth in new areas, voice changes, breast development in girls, broadening of shoulders in boys; these are driven by sex hormones.

- **Male reproductive system** — Testes (sperm production + testosterone), vas deferens, seminal vesicles + prostate (add fluid → semen), urethra/penis.
- **Female reproductive system** — Ovaries (egg + oestrogen/progesterone), fallopian tubes (site of fertilisation), uterus (embryo development), vagina.
- **Menstrual cycle** (~28 days) — One egg released roughly every 28 days. If not fertilised, the inner uterine lining is shed as **menstruation**. If fertilised in the fallopian tube, the zygote travels to the uterus and implants. The **placenta** develops to nourish the embryo with oxygen and nutrients from the mother's blood and remove its waste.

Reproductive health

Sexual contact can also transmit **diseases (STIs)** — bacterial (gonorrhoea, syphilis) and viral (HIV/AIDS, warts). Use of barrier methods (condoms) reduces this risk and prevents pregnancy. Other contraceptives:

- **Mechanical/Barrier:** condoms, diaphragms.
- **Chemical:** oral contraceptive pills (alter hormone levels).
- **IUCDs** (Copper-T): inserted in uterus; prevents implantation.
- **Surgical:** vasectomy in men, tubectomy in women — permanent.

Female foeticide — illegal in India — has skewed our sex ratio in many states; prenatal sex determination is banned by law (PCPNDT Act).

★ MUST-REMEMBER POINTS

- Asexual: 1 parent, identical offspring (clone). Sexual: 2 parents, gamete fusion, variation.
- Fission (Amoeba), Budding (Hydra/Yeast), Fragmentation (Spirogyra), Regeneration (Planaria), Spores (Rhizopus).
- Flower parts: Stamen (anther + filament) male; Pistil (stigma + style + ovary + ovule) female.
- Fertilisation in plants: pollen tube grows down style to ovule; zygote $\rightarrow$ embryo, ovule $\rightarrow$ seed, ovary $\rightarrow$ fruit.
- Site of fertilisation in humans: fallopian tube. Placenta nourishes embryo.
- Menstrual cycle ~28 days; menstruation = shedding uterine lining if no fertilisation.
- Contraceptive methods: barrier, chemical, IUCD, surgical. Condoms also prevent STIs.

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